

Machine Learning Capstone Project

Instructional Guidelines & Curriculum Framework

UGASS DATA SCIENCE CLUB X ZINDI

Submit final project requirements here: <https://forms.gle/JWi6h3iD76RJVxLx7>

Submission deadline: May 16, 2026 @12pm

1 Project Overview

Welcome to your Machine Learning Capstone! This project is the culmination of your learning journey, designed to transition you from guided tutorials to solving real-world business problems. Over the next week, you will work in teams of 5 to apply end-to-end machine learning workflows.

Learning Objectives:

- Execute a complete data science lifecycle from raw data to insights.
- Demonstrate proficiency in data cleaning, feature engineering, and exploratory analysis.
- Implement and compare multiple machine learning algorithms.
- Effectively communicate technical findings to both technical and non-technical audiences.

2. Project Options

Teams must choose **one** of three datasets provided. Note: No external sourcing of data is allowed for this capstone.

3. Project Requirements

Every project must include the following technical components:

- **Data Sourcing:** Use the provided CSV file. Perform initial checks (shape, info, head).
- **Exploratory Data Analysis (EDA):** Visualise distributions, correlation matrices, and relationships between features and the target. Identify outliers (If there are, explain how they are being handled)
- **Preprocessing:** Handle missing values, encode categorical variables (One-Hot/Label encoding), and scale numerical features (Standard/MinMax).
- **Feature Engineering:** Create at least 2 new features that might add predictive power to your model.
- **Modeling:** Train at least 3 different models (e.g., Logistic/Linear Regression, Random Forest, XGBoost).
- **Evaluation:**
 - *Classification:* Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
 - *Regression:* MAE, MSE, RMSE, and R^2 Score.
- **Hyperparameter Tuning:** Use GridSearchCV or RandomizedSearchCV to optimize your best-performing model.

4. Deliverables

Teams will submit two primary components at the end of the week.

A. GitHub Repository

The repository must be organized as follows:

- `/data`: The raw dataset (provided).
- `/notebooks`: Clearly numbered Jupyter Notebooks (e.g., `01_EDA.ipynb`, `02_Modeling.ipynb`).
- `/src`: Any helper `.py` scripts (optional).
- `README.md`: A professional project summary, setup instructions, and team member roles.
- `requirements.txt`: List of libraries used.

B. PowerPoint Presentation (10-12 Slides)

Your presentation should follow this structure:

1. **Title Slide:** Project Name & Team Member Names.
2. **Problem Statement:** What are you solving and why does it matter?
3. **EDA Highlights:** 3-4 key visual findings from the data.
4. **Data Preparation:** Steps taken to clean and transform the data.
5. **Model Selection:** Which algorithms were tried?
6. **Results:** Comparative table of model performance.
7. **The Winning Model:** Detailed look at your final model's performance.
8. **Feature Importance:** What drove the predictions?
9. **Business Recommendations:** Actionable insights based on your model.
10. **Challenges & Future Work:** What was hard? What would you do next?

5. Grading Rubric

Category	Criteria	Points
Data Preparation	Thorough EDA, handling of nulls/outliers, and meaningful feature engineering.	25
Model Building	Correct implementation of at least 3 models and proper hyperparameter tuning.	25
Evaluation	Correct use of metrics and interpretation of results (confusion matrix/residual plots).	20
Code & Repo	Clean, commented code and a well-organized GitHub repository.	15
Presentation	Clarity of communication, slide design, and ability to answer questions.	15
Total		100

6. Timeline & Milestones

Day 1: Project Kick-off. Team formation, dataset selection, and initial EDA. **Checkpoint:** Basic summary stats and 5 plots.

Day 2: Data Engineering. Cleaning, encoding, scaling, and feature engineering. **Checkpoint:** Clean dataset ready for modeling.

Day 3: Baseline Modeling. Implement 3 standard algorithms with default parameters. **Checkpoint:** Comparison table of baseline scores.

Day 4: Optimization. Hyperparameter tuning, cross-validation, and final model selection. Start drafting slides.

Day 5: Finalization. Finish GitHub README, polish the PowerPoint, and Present to the class.

7. Tools and Libraries

- **Language:** Python 3.x
- **Environment:** Jupyter Notebook / Google Colab
- **Data Manipulation:** Pandas, NumPy
- **Visualisation:** Matplotlib, Seaborn
- **Machine Learning:** Scikit-Learn
- **Advanced Modeling:** XGBoost or LightGBM (Optional)
- **Version Control:** Git & GitHub

8. Tips and Common Mistakes

Best Practices:

- **Divide and Conquer:** Assign roles (e.g., 1 Lead for EDA, 2 for Modeling, 1 for Repo/Cleaning, 1 for Presentation/Documentation).
- **Validation Strategy:** Always use a Train-Test split (80/20 or 70/30) before touching your models.
- **Document as you go:** Don't leave the README and slides for the last hour.

Pitfalls to Avoid:

- **Data Leakage:** Scaling or computing means on the *entire* dataset instead of just the training set.
- **Imbalance Ignored:** For classification, if 95% of data is 'No Churn', an accuracy of 95% is a "failed" model. Check your Recall/F1!
- **Overcomplicating:** Sometimes a simple Logistic Regression is better and more interpretable than a complex neural network.